

Rank Before You Merge: Stage-Consistent Visual Token Selection for Merger-Equipped Vision-Language Models

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Abstract—Visual-token selection around a native spatial merger is an operator-order problem: top- k decisions made before and after a lossy merger need not retain the same evidence. We introduce Rank-Before-Merge (RBM), a training-free rule that scores native 2×2 units on merger-input features at a fixed budget before applying the native merger. It adds no learned or query-dependent scorer. With iso-budget query-blind L2, RBM outperforms matched post-merger L2 on all nine text-dense cells across Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B, with gains of +11.0–+38.4 percentage points and +235–+432 OCRBench points. A full-split matched-depth Qwen3-VL control shows the same text-dense effect but a -5.64 percentage-point pre-final-minus-post difference on GQA (95% CI $[-6.42, -4.86]$), establishing a workload boundary rather than a universal stage preference. Mechanism probes show rank reshuffling and text-stroke demotion; a ranking-swap control recovers pre accuracy with identical kept sets. Query-conditioned FastV remains stronger in several reference-grounded and object-centric settings, so RBM is positioned as a robust OCR-oriented default. At 25% retention, compression is stage-neutral in speed, increasing throughput by 68% on Qwen3-VL and 1.8–2.5 $\times$ on InternVL3.

Index Terms—visual token compression, vision-language models, training-free inference, multimodal foundation models, inference efficiency

I. INTRODUCTION

Multimodal inference cost is a visual-token problem. Visual-language models (VLMs) process images as hundreds to tens of thousands of visual tokens, making quadratic attention and key-value memory major inference costs. Visual-token compression is therefore central to efficient document understanding, scene-text QA, and OCR. In merger-equipped architectures such as Qwen-VL and InternVL3, 2×2 patch groups are first merged, while prior work focuses mainly on which tokens to keep and how to score them. The selection stage, before or after the native merger, is rarely isolated as a matched variable. Prior work notes Qwen’s native token reduction [1] and OCR degradation from post-encoder merging [2], [3]; we directly test the boundary by scoring the same units with query-blind L2 on merger-input (pre) or merger-output (post) features at the same per-image budget.

Stage-consistent selection: Rank-Before-Merge (RBM). RBM resolves the operator-order ambiguity by scoring each merger-input unit, retaining the top- κ , and applying the unmodified native merger only to the survivors. Its method contract fixes the unit granularity, per-image budget, shared mask, and native visual interface. Because merging is per-unit, a fixed kept set receives the same native operation; byte identity

is verified on Qwen3-VL. The method therefore targets merger-induced *ranking* changes without adding a learned scorer, query input, or LLM-attention probe (§III-C).

What falls out of it — a stage-consistent effect across models. Under iso-model, iso-budget, and a shared query-blind L2-family rule, pre leads post by 11.0–38.4 pp on official text-dense splits across two Qwen generations and by 34.6–37.4 pp plus 432 OCRBench points on InternVL3, whose merger design differs. A full-split matched-depth pure-stage control retains the text-dense effect but yields -5.64 pp on Qwen3-VL GQA (95% CI $[-6.42, -4.86]$); the result is workload-conditional and widens with compression depth (§IV).

Why the stage matters. On Qwen3-VL, the learned merger weakly preserves unit ranks (Spearman $\rho = 0.14$ – 0.36), preferentially demotes high-edge text-stroke units on documents, and yields to a ranking-swap control: the post path recovers pre accuracy with identical selected-unit sets (Jaccard = 1.000 in four model $\times$ benchmark cells). Thus, where tested byte-exactly, the ranking rather than the forward path explains the gap (§V). This matched-rule result is distinct from the query-conditioned FastV comparison: FastV remains stronger on several scene-text/object-centric cells, so RBM is a query-blind, OCR-preserving robust default, not a universal winner (§IV-D).

Contributions. First, we formulate representation-dependent top- k selection around a native merger as a selection–merger ordering problem and state its kept-set equivalence condition. Second, we derive RBM as a stage-consistent design with an explicit structural contract and no learned parameters (§III). Third, we establish a workload-conditioned stage effect across three models and two merger designs, identify rank reshuffling and text-stroke demotion as the Qwen3-VL mechanism, and map the boundary against FastV and five prespecified extensions (§IV–VI).

II. RELATED WORK

Token pruning and merging in VLMs. Most methods score visual tokens and drop or merge the least important before or within the early LLM layers. **FastV** prunes at the second LLM layer from the attention visual tokens receive [4]; it is our primary same-model, query-conditioned strong baseline — the method that is stronger on several question-conditioned non-OCR cells but weaker in the tested OCR regime (§IV-D). **PyramidDrop** drops tokens progressively across LLM layers [5]; **FasterVLM** accelerates attention-based pruning [6];

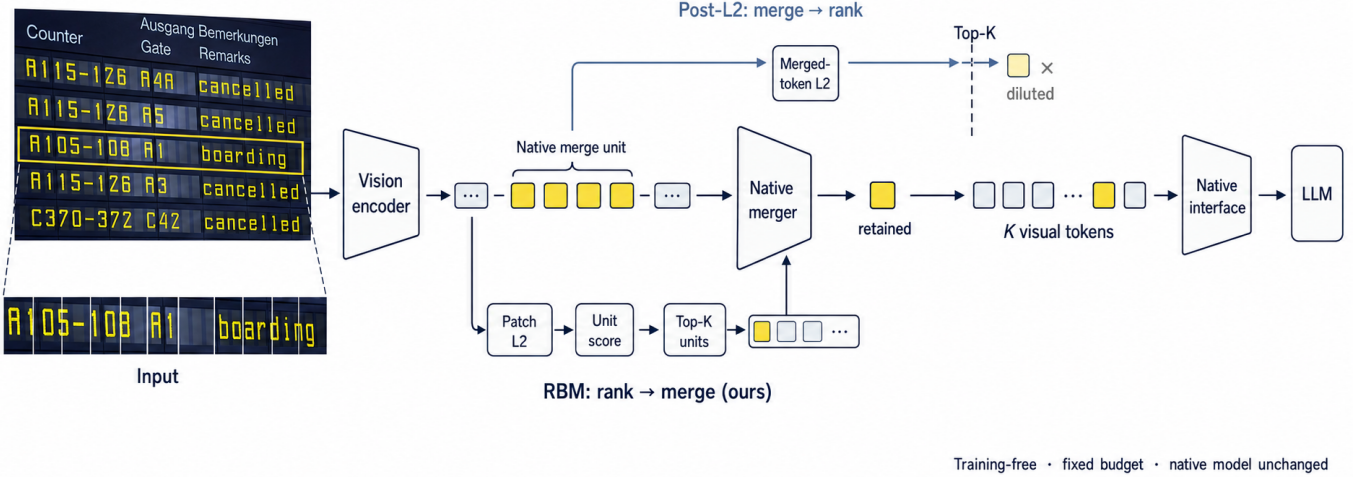


Fig. 1. **Rank-Before-Merge (RBM) method overview.** The selected OCRBench example contrasts conventional Post-L2, which first applies the native merger and then ranks merged-token features, with RBM, which scores patch features within native 2×2 merge units, keeps the top- K units, and only then invokes the unmodified native merger. Both paths use the same visual-token budget. RBM is training-free and preserves the native visual interface and downstream language model; the highlighted input region illustrates answer-bearing evidence that can be diluted by post-merger ranking.

SparseVLM explores query-conditioned sparsification [7]; **PruMerge** merges by attention-derived importance [8]; **FitPrune** fits lightweight importance predictors [9]. Other training-free selectors use visual [CLS] attention, graph propagation, or multi-cue suppression [10]–[12], while unified recent systems combine importance with diversity or text-guided merging [13]–[15]. These methods prune or merge at different encoder-output or LLM-side hook points. Among the methods reviewed here, we found no matched comparison of selection immediately before versus after a model’s native spatial merger under iso-model, iso-token, and a shared query-blind L2-family rule.

Token-merging lineage (ToMe → AdaptMerge). Token merging originates with **ToMe** [16] (bipartite soft matching, post-encoder, query-blind); **AdaptMerge** [2] adapts it to large multimodal models and reports that language-guided merging closes most of the OCR gap plain ToMe incurs — corroborating evidence for our lossy-merger thesis, cited as support, not as a competitor. This lineage adds a *second*, learned merging stage; we hold the native merger fixed and ask only whether selection should precede or follow it, so it is methodologically distinct from our stage-axis test. Learned visual projectors such as TokenPacker condense tokens while preserving fine local cues [17]; unlike our inference-time test, they alter the projector architecture and require model adaptation.

VisionZip and Qwen-family compression. **VisionZip** selects “dominant” tokens by CLS-to-patch attention and merges the remainder into “contextual” tokens [3]. In the official Qwen2.5-VL implementation we inspected, selection operates on embeddings after the PatchMerger, i.e. on the post-merger side of our axis. Its authors’ own Qwen2.5-VL numbers are a model/stage-mismatched anchor only: OCRBench 81.5 $\rightarrow$ 70.5 at 50% retention while general tasks hold, with the README caveat that gains are “less striking” because the merger already

compresses — the same effect seen from the post-merger side; we claim no head-to-head victory over it (§IV-D). Qwen-specific methods **GlimpsePrune** [18] and **VScan** [19] provide additional Qwen-family compression comparisons without the matched stage-only control used here. The concurrent **AnchorPrune** study applies Qwen2.5-VL selection after its model-specific spatial processing and token merging [20], again placing it on the post-merger side rather than matching both sides of the boundary. Recent layerwise methods instead score token evolution: TransPrune uses token-transition variation [21], while V2Drop exploits variation across adjacent layers [22]. They are complementary to the controlled native-merger stage axis studied here.

The 2026 frontier. Recent work broadens token reduction beyond scalar attention ranks through hierarchical selection [23]. Set coverage, reconstruction error, and functional sensitivity motivate joint or projector-aware selection [24]–[28]; learned or strongly conditioned compressors instead model semantic labels, information flow, or head importance [29]–[32]. Other work makes token lifetime explicitly layer-dependent [33]–[35], or preserves discarded content through object-centric, optimal-transport, or hybrid aggregation [36]–[38]. Document- and OCR-specific studies further show that generic accuracy can hide lost textual evidence [39], [40]. These advances reinforce the need to localize the intervention stage; unlike these studies, our experiment directly matches the two sides of a frozen model’s native spatial merger.

The design point we isolate. We do not claim a topological vacuum. QuietPrune-style in-ViT pruning [41] is a different, earlier, trained stage; PixelPrune removes redundant pixel patches before the ViT [42]; FastV/PyramidDrop select inside the LLM; VisionZip selects on the opposite (post-merger) side. The combination we isolate is narrower: on a merger-equipped

VLM, leaving the native merger untouched, scoring the merger-*input* 2×2 units by a query-blind saliency norm, keeping the top- κ , and handing only the survivors to the unmodified native merger — together with Qwen3-VL mechanism evidence (M1–M3) that is not evaluated in our cited comparisons and explains why this stage can dominate on text-dense workloads.

Efficiency evaluation of VLMs. Reported speedups depend heavily on engine and protocol. We adopt the lmms-eval family of *official* scorers for accuracy [43]; EffiVLM-Bench independently emphasizes cross-backbone, multi-metric acceleration evaluation [44]. We confine all throughput claims to a single engine (§VII).

III. METHOD — STAGE-CONSISTENT RANK-BEFORE-MERGE

A. Problem formulation: selection–merger ordering

Let $U = \{u_i\}_{i=1}^N$ be the native merge units and let M denote the model’s per-unit merger. A representation-dependent scorer induces $a_i = s_{\text{pre}}(u_i)$ before merging and $b_i = s_{\text{post}}(M(u_i))$ after merging. At budget k , the two decisions are

$$K_{\text{pre}} = \text{TopK}_k\{a_i\}_{i=1}^N, \quad K_{\text{post}} = \text{TopK}_k\{b_i\}_{i=1}^N. \quad (1)$$

Their visual outputs are $\{M(u_i) : i \in K_{\text{pre}}\}$ and $\{M(u_i) : i \in K_{\text{post}}\}$, respectively. For a *fixed* mask, selection and a per-unit merger are compatible; the non-equivalence arises because the top- k mask itself depends on whether saliency is measured before or after M .

Selection-equivalence condition. With deterministic tie breaking, the two stages are equivalent at budget k exactly when $K_{\text{pre}} = K_{\text{post}}$. A sufficient condition for equivalence at all budgets is $b_i = g(a_i)$ for a strictly increasing g . We quantify violation by the kept-set disagreement

$$D_k = 1 - \frac{|K_{\text{pre}} \cap K_{\text{post}}|}{|K_{\text{pre}} \cup K_{\text{post}}|}, \quad (2)$$

the complement of kept-set Jaccard. Section V tests the rank-preservation premise and then isolates whether changed unit identity, rather than a changed forward path, accounts for the measured Qwen3-VL gap.

Architectural setting.

We study three merger-equipped VLMs with two merger designs. Qwen3-VL-8B and Qwen2.5-VL-7B use a learned PatchMerger [1], [45]; InternVL3-8B uses pixel-shuffle down-sampling [46]. All are served in bf16, eager attention, vLLM 0.19 (V1) [47], single A40 (46 GB), greedy decoding. The shared pipeline is: image $\rightarrow$ ViT patch tokens $\rightarrow$ native merger (2×2 groups $\rightarrow$ one token) $\rightarrow$ language model. The Qwen merger is *not* an averaging pool: the four patch features of each 2×2 group are concatenated and projected by a small learned MLP (LayerNorm $\rightarrow$ fc1 $\rightarrow$ GELU $\rightarrow$ fc2) to one unit vector; the lossy step is the four-to-one nonlinear projection, so a kept unit’s representation is a nonlinear function of its four patches. We verified this against the served implementation on both Qwen generations.

The stage axis (the experimental variable). The two arms share model, token count, decoding, and a query-blind L2 magnitude rule, while evaluating that rule on the native feature state available on each side of the merger:

- **Pre-merger (RBM).** For each 2×2 unit, average the L2 norms of its four merger-input patch vectors, keep the top- κN , and pass only the survivors through the unmodified native merger (and, for Qwen3-VL, each deepstack merger). The merger operates exactly as in the uncompressed model, on a subset of units.
- **Post stage (contrast).** Either score the N *merged* units on merger-*output* features and keep the top- κN (the stage used by published compression methods for merger-equipped VLMs, including VisionZip’s Qwen path), or — the stronger, query-conditioned contrast — prune inside the LLM by layer-2 attention (the FastV stage). The main matrix (Table I) uses the post-merger-L2 contrast from the same query-blind L2 family; the layer-2 FastV contrast is used only in Table II.

RBM contract. The method (i) masks only complete native merge units, (ii) retains exactly the per-image top- k budget, (iii) shares unit identity across the main and deepstack streams, and (iv) leaves the native merger and downstream visual interface unchanged. These constraints separate a stage intervention from changes in token count, merger architecture, or query access. Figure 2 visualizes how the two matched-budget stages retain different spatial evidence on representative TextVQA and DocVQA inputs.

Budget definition. Retention κ is defined over *merge units* relative to each image’s **own** full unit count, so two methods at the same κ feed the language model the same number of visual tokens per image; we verify equality of the mean post-merger token count per benchmark (**iso-token** control), so any accuracy difference is attributable to *which* units survive and the representational state on which selection is made, not to token count. We report $\kappa \in \{0.25, 0.125\}$ (25% / 12.5% retention); per-benchmark token counts are printed with each table.

B. The L2 selector as a control variable

We deliberately use the simplest **text-agnostic, query-blind** L2-magnitude family. For unit u with four merger-input patch vectors $h_{u,1:4}$ and native merger M , the implemented scores are

$$s_{\text{pre}}(u) = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2, \quad s_{\text{post}}(u) = \|M(h_{u,1:4})\|_2.$$

Thus the ranking functional is matched at the level of feature magnitude, while its representation is the experimental stage: four input vectors before the nonlinear merger versus its output vector after it. For Qwen3-VL, the headline operational pre arm takes h from the first deepstack tap (ViT layer 8), whereas the matched-depth pre-final control takes h from the final ViT block at the main-merger boundary (§IV-B). We freeze plain RBM and add no deployed variant. Mechanism-side probes in §V-D and §VI test a variance-weighted score and a second scorer family; they bound generality but do not redefine the method. Unlike the merging lineage of §II, RBM introduces no second learned merger: it manipulates only the *set* of units the native merger sees, making pre vs post a stage intervention rather than a scorer comparison.

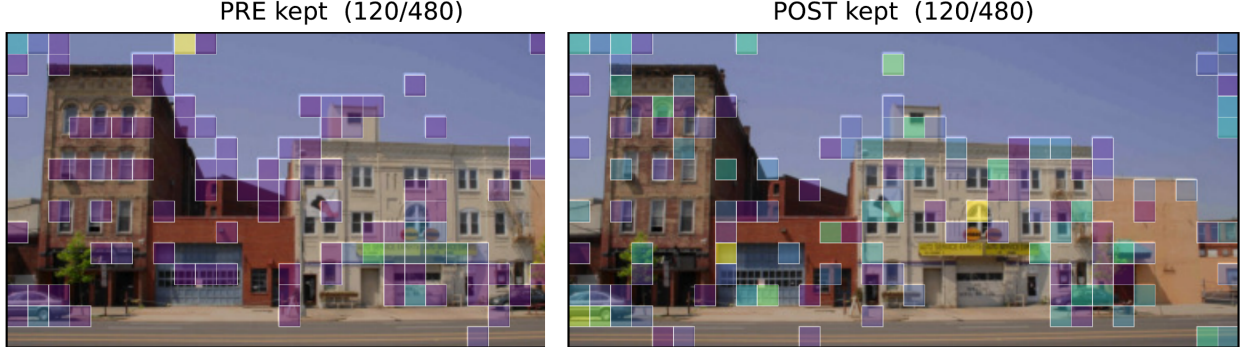


Fig. 2. **Representative visual comparison at 25% retention.** The same TextVQA storefront is scored with query-blind L2 before and after Qwen3-VL’s native merger. The pre- and post-merger selectors retain the same budget but visibly different spatial units, especially around the text-bearing facade. Sample identifiers, prompts, score-source details, and the companion document example are retained in Supplementary S2 and the artifact package.

C. Rank-Before-Merge

Given an image with N merge units, compute $s(u)$ on merger-input features for every unit, retain the top $k = \kappa N$, and invoke the native merger(s) on the retained units only. Unit identity is shared across the main and deepstack streams (deepstack mergers process intermediate-depth features of the *same* units), so pruning a unit removes it from every stream. No attention weights are required and no parameters are trained.

Selection cost and deployment. For d -dimensional patch features, L2 scoring costs $O(Nd)$, a heap-based top- k costs $O(N \log k)$, and storing the scores costs $O(N)$. RBM does not reduce the ViT computation before its tap; it reduces the units entering the native merger and language-model interface. The method adds zero trainable parameters and no query- or LLM-attention pass; end-to-end stage neutrality is therefore evaluated empirically in §VII, rather than inferred from asymptotic notation.

Unit equivalence (the structural invariant). Because the mask is at *unit* granularity and the merger is per-unit, a fixed kept set supplies the same four patches to the same native per-unit operation. Selection cannot prevent intra-unit combination; it changes which units reach the merger. The resulting outputs are byte-identical in the verified Qwen3-VL path; other families are limited to the evidence stated in §VIII. Pre-merger selection keeps units on their raw (merger-input) saliency; post-stage selection keeps them on a saliency the merger has already rewritten (§V). This invariant is what makes the ranking-swap control of §V-C decisive.

Post-merger selection (the contrast): replace lines 3–5 of Algorithm 1 by $U' \leftarrow \text{NativeMerger}(U)$; $\text{score}'[u'] \leftarrow \|f_{\text{out}}(u')\|_2$ on merger-output features; $K' \leftarrow \text{top-}k$ by score' ; $\text{keep}(K')$. Everything downstream is identical.

Configuration details. Per-family tap points (Qwen3-VL deepstack[0] at ViT layer 8, Qwen2.5-VL main merger input, InternVL3 pixel-shuffle input) and the Qwen2.5-VL position-cursor fix are detailed in Supplementary §S8, including the cross-generation configuration diagnostic. Supplementary S1 summarizes the corresponding rank-agreement distributions; Figure 3 shows the retention-depth response.

Algorithm 1 Rank-Before-Merge (RBM), one image

Require: image x ; retention κ ; query-blind scorer $s = \|\cdot\|_2$

Ensure: compressed visual token sequence z

- 1: $P \leftarrow \text{ViT patch tokens of } x \text{ \{16 px footprint\}}$
 - 2: $U \leftarrow \text{group } P \text{ into } N \text{ native } 2 \times 2 \text{ units \{32 px footprint\}}$
 - 3: **for** each unit $u \in U$ **do**
 - 4: $\text{score}[u] \leftarrow \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2$ {mean patch L2}
 - 5: **end for**
 - 6: $k \leftarrow \max(1, \text{round}(\kappa N))$; $K \leftarrow \text{topk}(\text{score}, k)$
 - 7: $z_{\text{main}} \leftarrow \text{NativeMerger}(\{u : u \in K\})$ {merge survivors only}
 - 8: $z_{\text{ds}} \leftarrow \text{DeepstackMergers}(\{u : u \in K\})$ {Qwen3-VL only}
 - 9: $z \leftarrow \text{NativeVisualInterface}(z_{\text{main}}, z_{\text{ds}})$
 - 10: **return** z {feed standard LLM}
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IV. MAIN EXPERIMENTS

A. Setup

Benchmarks and metrics. We use TextVQA VQA accuracy [48], DocVQA ANLS [49], the official OCR-Bench score [50], and GQA normalized exact match [51]. Official split sizes are 5000, 5349, 1000, and 12578, respectively. Scorers are verbatim ports of the lmms-eval family [43], with ground-truth self-test passing 200/200.

Engines and fairness protocol. RBM and the none / post cells run in vLLM; the FastV baseline runs in an independent HuggingFace transformers eager harness (it has no vLLM path). Accuracy is comparable across engines (the harness bit-degrades to native at zero drop, 8/8 per-sample identical; HF-vs-vLLM none cells agree 16/16); **throughput is not** — all efficiency numbers are confined to vLLM (§VII). Fairness is enforced by the same keep ratio relative to each image’s own token count and by reporting mean post-merger token counts.

Significance. Each pre-vs-post Δ pairs the same samples under greedy decoding. The prespecified primary family comprises the nine model $\times$ text-dense-benchmark pre-vs-post contrasts at 25% retention in Table I. We use 20,000-resample paired-bootstrap confidence intervals on per-item metric differ-

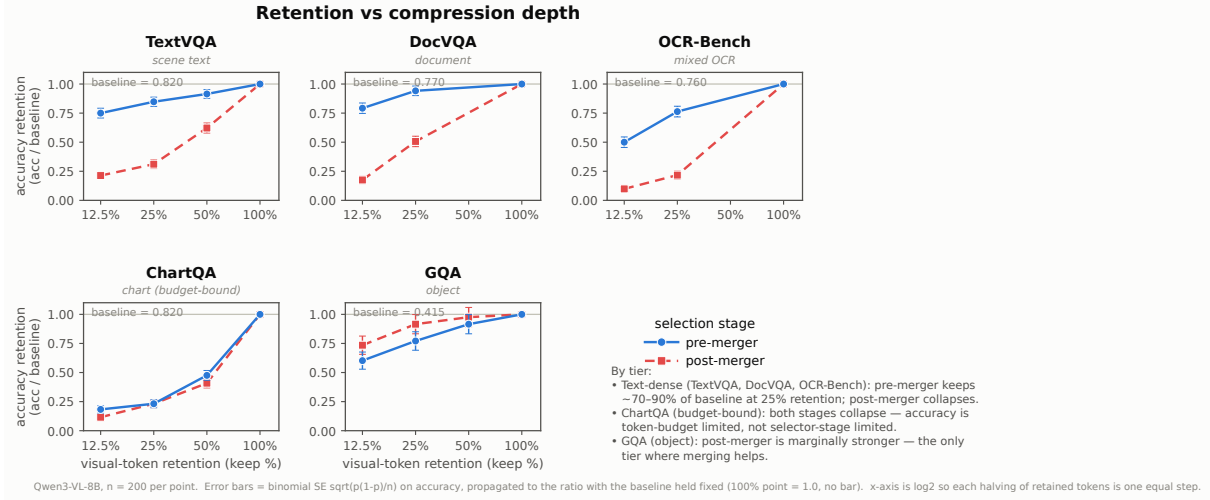


Fig. 3. **Accuracy response to compression depth.** Normalized accuracy is shown on Qwen3-VL-8B ($n = 200$ per point) as the visual-token budget decreases. Pre-merger selection preserves substantially more text-dense accuracy, while ChartQA is primarily budget-limited and the small-sample GQA curve is not used to claim a universal stage preference. The full-split matched-boundary GQA control is reported in Section IV-B. Rank-agreement distributions and their full table are retained in Supplementary S1, avoiding duplication with Fig. 5.

ences and two-sided paired sign-flip permutation tests; Holm correction across these nine tests gives $p_{\text{Holm}} \leq 4.5 \times 10^{-4}$ for every contrast. McNemar is reported only as a binary-outcome cross-check for OCRBench containment and GQA exact match. Other budgets, GQA, mechanism probes, FastV comparisons, and extension gates are secondary or exploratory and are not pooled into the primary family. The metric-level tests preserve partial credit in VQA accuracy and ANLS rather than binarizing those metrics.

B. Main result across three models — official full splits (Table I)

The stage-consistent effect holds on both generations. Every text-dense Δ_{25} is large and significant: RBM leads post-merger selection by +38.4 pp (TextVQA), +24.3 pp (DocVQA), +363 pts (OCR-Bench; 547 vs 184, 3.0 $\times$) on Qwen3-VL, and by +26.1 pp, +11.0 pp, +298 pts (480 vs 182, 2.6 $\times$) on Qwen2.5-VL; all paired-bootstrap 95% CIs exclude zero. The gap is robust to deeper compression: at $\kappa = 0.125$ the post stage degrades much faster (Qwen3-VL OCR-Bench 184 $\rightarrow$ 53 vs RBM 547 $\rightarrow$ 350, widening the pre/post ratio from 3.0 $\times$ to 6.6 $\times$), and all six text-dense $\Delta_{12.5}$ remain significant under paired metric-level inference.

A full-split matched-boundary control (pre-final). The original pre arm on Qwen3-VL ranks at the deepstack[0]-input tap (layer 8, far upstream) and slices all four mergers, while the post arm ranks at the main-merger output — so the headline contrast differs in *both* stage and feature space. To isolate the stage at a matched boundary, a **pre-final** control ranks L2 at the main-merger’s *own* input (final ViT-block features) and selects the top- κ merge-units from the visual output, leaving the deepstack mergers untouched. On the full official splits ($n = 5000/5349/1000/12578$), pre-final and post use identical sample IDs and per-sample token counts. The only intended difference is *before versus after* the lossy 2×2 main merger — a pure stage variable.

Pre-final beats post by +27.68 pp on TextVQA, +4.59 pp on DocVQA, and +235 OCRBench points; all three paired 95% CIs exclude zero. On GQA, pre-final is 0.4207 versus post 0.4771, a -5.64 pp difference (95% CI $[-6.42, -4.86]$, $n = 12578$). The full control table and paired uncertainty are in Supplementary S7b. The result therefore confirms a genuine text-dense before-merger advantage at the matched boundary, while GQA favors post-merger selection; the stage effect is workload-conditioned, not a universal ranking rule.

Object-centric GQA — a workload boundary, not a universal stage preference. The corrected full-split matched-boundary control favors post-merger selection on GQA (0.4207 pre-final versus 0.4771 post; -5.64 pp, 95% CI $[-6.42, -4.86]$, $n = 12578$). The original operational layer-8 contrast also favors post (-2.8 pp Qwen3-VL and -2.6 pp Qwen2.5-VL), while the layer-8 RBM arm partly compensates for the Qwen3-VL loss relative to pre-final (0.449 versus 0.4207). Thus the GQA result is not a true reversal that disappears at the matched boundary; it is a genuine but much smaller post-stage advantage on object-QA. Its magnitude is about one-fifth to one-sixth of the text-dense deltas, and no text-dense benchmark shows a crossover at the native-resolution full splits. We therefore state the conclusion as **workload-conditioned**, without claiming equivalence or universal pre-stage superiority.

A no-merger negative control (LLaVA-1.5). LLaVA-1.5-7B has a per-token MLP projector (Linear-GELU-Linear, no 2×2 spatial averaging). Running the same pre-vs-post L2 protocol (pre-MLP vs post-MLP ranking, keep 25%, $n=200$, greedy) yields *no* significant stage effect: TextVQA $\Delta = -2.17$ pp (McNemar $p=0.125$; none 0.452, pre 0.087, post 0.108), GQA $\Delta = -0.5$ pp ($p=1.0$; none 0.575, pre 0.375, post 0.380) — versus the merger-equipped Qwen3-VL’s +38.4 pp ($p \approx 5 \times 10^{-5}$) on TextVQA. In this no-merger control the trend is slightly reversed (post $\geq$ pre, n.s.). Together with the merger-equipped results, this is consistent with the spatial merger being a key condition for the large text-dense pre $>$ post gap. A single

TABLE I

OFFICIAL FULL-SPLIT METRICS ($n = 5000/5349/1000/12578$), GREEDY, ACROSS QWEN3-VL-8B, QWEN2.5-VL-7B, AND INTERNVL3-8B. RBM = PRE-MERGER L2 (OURS); POST = POST-MERGER L2 AT THE SAME TOKEN BUDGET. $\Delta_{25} = \text{PRE} - \text{POST}$ AT $\kappa = 0.25$. $\dagger$ MARKS THE ONLY SIGNIFICANT POST-STAGE LEADS (GQA); THESE USE THE DEEPSTACK[0]-INPUT (LAYER-8) PRE RANKING. THE SEPARATE FULL-SPLIT MATCHED-BOUNDARY QWEN3-VL CONTROL IN §IV-B ALSO FAVORS POST ON GQA (-5.64 pp FOR PRE-FINAL MINUS POST), SO THE RESULT IS A GENUINE WORKLOAD BOUNDARY RATHER THAN A CLAIM OF UNIVERSAL PRE-STAGE SUPERIORITY. INTERNVL3 USES A DIFFERENT MERGER AND FAMILY-SPECIFIC IMAGING CONFIGURATION; COMPARE WITHIN-MODEL DELTAS, NOT CROSS-MODEL ABSOLUTE VALUES. FULL PROTOCOL AND SKIP ACCOUNTING ARE IN THE SUPPLEMENTARY MATERIAL.

Model	Benchmark (metric)	none	RBM @25%	post @25%	Δ_{25}	RBM @12.5%	post @12.5%
Qwen3-VL-8B	TextVQA (VQA-acc)	0.844	0.605	0.222	+38.4 pp	0.472	0.132
	DocVQA (ANLS)	0.956	0.481	0.238	+24.3 pp	0.352	0.103
	OCR-Bench (/1000)	760	547	184	+363 pts	350	53
	GQA (exact match)	0.616	0.449	0.477	-2.8 pp $\dagger$	—	—
Qwen2.5-VL-7B	TextVQA (VQA-acc)	0.862	0.702	0.442	+26.1 pp	0.597	0.319
	DocVQA (ANLS)	0.949	0.636	0.526	+11.0 pp	0.455	0.245
	OCR-Bench (/1000)	817	480	182	+298 pts	328	70
	GQA (exact match)	0.604	0.559	0.585	-2.6 pp $\dagger$	—	—
InternVL3-8B	TextVQA (VQA-acc)	0.8338	0.7890	0.4148	+37.4 pp	0.7230	0.3064
	DocVQA (ANLS)	0.9221	0.7284	0.3820	+34.6 pp	0.5054	0.2451
	OCR-Bench (/1000)	852	753	321	+432 pts	—	—
	GQA (exact match)	0.6293	0.5993	0.6031	-0.4 pp (n.s.)	—	—

Table notes. (a) Qwen3-VL none DocVQA was re-run at max-model-len 49,152 and completed with 0/5349 skips (ANLS 0.956); the pre/post cells it anchors ran at the same native-resolution configuration and are directly comparable to each other. (b) OCR-Bench none cells skip 18 (Qwen3-VL) / 24 (Qwen2.5-VL) of 1000 long-OCR images on context overrun, scored 0 (conservative; compressed cells skip ≤ 5), so the uncompressed anchor is understated and the relative gaps are, if anything, conservative. (c) GQA was evaluated at $\kappa = 0.25$ only on the full split. (d) Iso-token control: at each model $\times$ benchmark, pre and post have identical mean post-merger token counts. The Qwen2.5-VL OCR-Bench cells at both budgets use the matched 4 M-pixel cap (mean tokens 229.6 and 131.5, respectively). (e) Primary inference uses paired-bootstrap 95% CIs and paired sign-flip permutation tests on the official per-item metric. All Qwen text-dense intervals exclude zero; Figure 4 shows the intervals. McNemar is additional for binary OCRBench/GQA outcomes. (f) InternVL3 uses pixel-shuffle downsampling and a 4 M-pixel DocVQA cap; its pre/post prompt token counts are identical within every cell. Its GQA -0.4 pp difference is indistinguishable: the paired 95% CI $[-0.93, +0.17]$ crosses 0 (permutation $p = 0.18$), so we claim no GQA stage lead on InternVL3; no preregistered equivalence bound exists, so we do not call it a statistical tie.

negative-control architecture cannot establish necessity or an if-and-only-if law.

C. Replication on a third family — InternVL3-8B (Table 1)

To test whether the Qwen direction replicates beyond that family, we run RBM and the post stage on **InternVL3-8B**, whose merger is pixel-shuffle downsampling — a different design from Qwen’s PatchMerger. The post arm here is **post-merger L2 top-k** — the same query-blind L2-magnitude family as the pre arm, at an identical token budget (ptid equal across arms), so the InternVL3 block of Table I isolates the before/after merger stage under iso-budget control at 25% retention. The query-conditioned in-layer FastV contrast is a separate campaign, reported in Table II.

Configuration boundary. InternVL3 DocVQA uses a 4 M-pixel cap, while the Qwen same-scope baseline cells of §IV-D use 600k; cross-model absolute values are not comparable. At 25% retention RBM costs -19.4 pp vs none on InternVL3 DocVQA and $-99/1000$ on OCR-Bench, versus only -4.5 pp on TextVQA and -3.0 pp on GQA. The claim is therefore the large within-model pre $>$ post margin, not losslessness on every text-dense workload.

At 12.5% retention ($r = 0.875$), TextVQA is 0.7230 pre vs post 0.3064 (**+41.7 pp**); DocVQA pre 0.5054 vs post 0.2451 (**+26.0 pp**) — the gap *widens* under deeper compression, as on Qwen.

Cross-family directional replication. InternVL3’s pre–post text-dense gaps ($+37.4/ +34.6$ pp / $+432$ pts) are the same

order and direction as Qwen3-VL’s ($+38.4/ +24.3/ +363$ pts) on all three text-dense benchmarks, and GQA stays within the red line — indistinguishable here (-0.4 pp), whereas Qwen3-VL shows a smaller but significant post-stage advantage in the full-split matched-boundary control (-5.64 pp for pre-final minus post; §IV-B). There is no text-dense crossover at either family’s native imaging configuration. These accuracy-level replications show that the shared L2-magnitude rule can favor the pre-merger side on a third model family with a different merger. They support — but do not by themselves establish — the Qwen3-VL selection-level mechanism as a cross-family explanation.

The ranking-swap comparison on InternVL3. The M3-style control (§V) — run the post-merger forward path but select units with the pre-merger ranking — closely matches pre-merger accuracy on the three tested InternVL3 cells: swap vs pre is $+0.00$ pp (TextVQA), $+0.11$ pp (DocVQA), and -0.50 pp (GQA), with kept-set Jaccard(pre,swap) = 1.0 and per-sample answer-identity 0.98–1.0 ($n = 200$, iso-token, no swap fallback). This supports a ranking-level explanation on a second merger architecture (pixel-shuffle, not just Qwen’s PatchMerger), but the nonzero residuals mean that the byte-exact causal attribution remains specific to Qwen3-VL.

D. A workload-conditioned regime map — RBM vs the query-conditioned strong baseline FastV (Table 2)

FastV is the query-conditioned strong baseline in this comparison — it selects inside the LLM after cross-attention has mixed

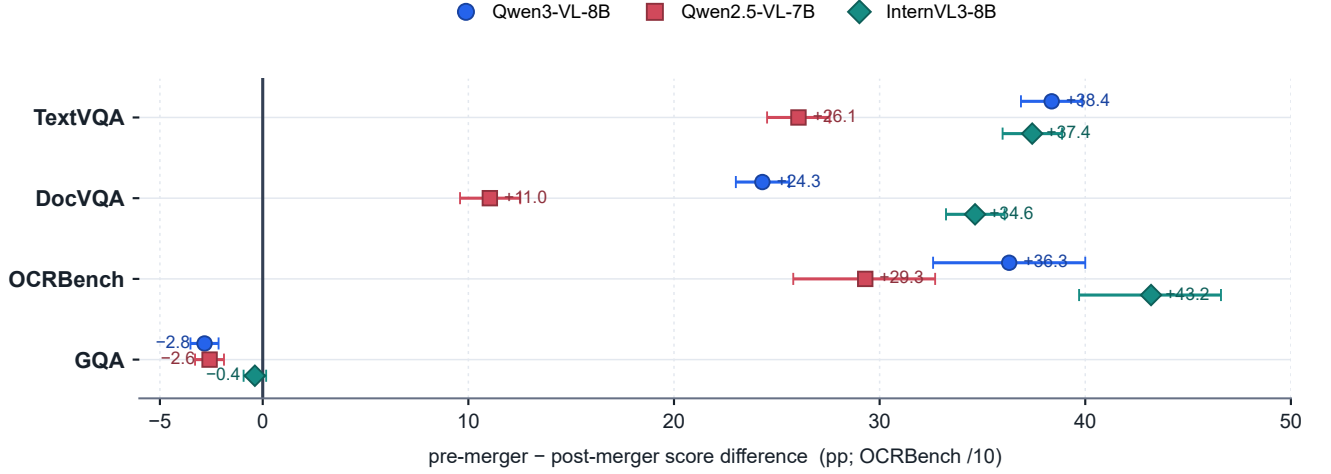


Fig. 4. **The text-dense stage effect replicates across models and merger designs.** Points are paired pre–post metric differences at 25% retention; whiskers are paired-bootstrap 95% confidence intervals on the official full splits. OCRBench’s /1000 score is divided by 10 for a common percentage-point axis. Qwen uses PatchMerger; InternVL3 uses pixel shuffle. GQA bounds the result: Qwen’s original layer-8 pre-ranking favors post, while InternVL3’s interval crosses zero. Values correspond to Table I.

in the question — and we give it every advantage: we compare RBM against FastV at its **best layer** K ($K = 3$; a K -sensitivity probe found $K = 3$ best), reported at identical n /scope with paired skips and official rescoring, so the comparison cannot be attributed to a handicapped K or a mismatched sample set. We read the result as a **workload-conditioned regime map** — where query-conditioned selection wins, where it fails, and why — not as a single-winner race. At its canonical progressive schedule PyramidDrop is near-lossless but occupies a different budget point (equivalent mean retention $0.625 \approx 2.1 \times$ ours); forced to an honest iso-25% schedule it collapses (TextVQA 0.073 / GQA 0.305 / OCR 0.005 on Qwen3-VL), whereas RBM holds (0.60 / 0.42 / 0.58). There is therefore no universal winner.

Full-split OCR confirmation. To check that the OCR advantage is not an artifact of the small native-resolution probe, we additionally evaluated all 1000 OCRBench items under the same HF eager harness and a 4 M-pixel cap. RBM scores 559 versus FastV’s 413 on Qwen3-VL and 382 versus 295 on Qwen2.5-VL. The gaps are +14.6 and +8.7 percentage points, respectively, with symmetric skips across arms. These full-split rows confirm the direction at larger scale; because their pixel cap differs from the older native-resolution $n = 200$ rows, they are reported as confirmation rather than silently substituted for the regime-map cells.

Workload-conditioned regime map. FastV leads several question-conditioned cells. FastV-k3 leads RBM on Qwen3-VL TextVQA and GQA (+17.2 / + 8.9 pp), while the Qwen3-VL DocVQA comparison is statistically indistinguishable (RBM 0.5924 vs FastV 0.5863; paired difference inconclusive). FastV also leads on Qwen2.5-VL TextVQA (+7.8 pp). This is the query-dependent headroom made reachable: *which* region matters for *this question* becomes visible only once cross-attention has seen the question, exactly where FastV operates and exactly where the prespecified query-level gate of §VI

could not reach before the merger. **RBM leads the tested OCR regime.** RBM wins OCR-Bench on **both** families (+16.0 pp Qwen3-VL, +8.5 pp Qwen2.5-VL); the Qwen2.5-VL $n = 200$ GQA/DocVQA cells favor RBM by +4.5 / + 2.1 pp but their paired 95% CIs cross 0 ($[-10.5, +1.5]$ / $[-10.1, +5.8]$), so they are exploratory/inconclusive at $n = 200$, not statistically robust RBM edges. The observed OCR split persists at FastV’s best tested K ; within this comparison, later query conditioning does not recover the OCR accuracy retained by RBM. **The map is the two failure modes of §V-D.** Documents = selection misdirection — the merger’s reshuffle misleads selection made after it away from text, so the OCR regime is lost by the selectors that tap the stream post-merger; scene text = feature degradation under a budget every 25%-keep method pays, and object scenes = near-null reshuffle — together the low-OCR regime where the query-dependent headroom lives. We report FastV’s wins honestly: FastV is a strong query-conditioned baseline that wins its regime; RBM is the robust default (never collapses, uniquely holds dense OCR), not a method that leads query-conditioned pruning everywhere.

V. MECHANISM — WHY THE MERGER IS LOSSY

We give the pre-vs-post gap a mechanistic account in three parts (M1–M3), measured on **Qwen3-VL-8B** (M1/M2 on a deterministic seed-0 $n = 64$ sample per benchmark; M3 on $n = 200$ with the same invocation as the headline cells), then state its architectural scope.

A. M1 — the merger reshuffles unit ranks

Per image we correlate the *pre* ranking (merger-input unit L2) with the *post* ranking (merged-token L2) over all units. The two rankings are barely related, and the reshuffle tracks text density: Spearman $\rho = 0.137$ (DocVQA, text-dense document)

TABLE II

WORKLOAD-CONDITIONED REGIME MAP — SAME-SCOPE PAIRED COMPARISON OF FASTV AT BEST $K = 3$ (THE QUERY-CONDITIONED STRONG BASELINE) VS RBM (PRE, THE QUERY-BLIND ROBUST DEFAULT) @25% RETENTION, BOTH QWEN MODELS, OFFICIAL RESCORING. CELLS ARE PAIRED WITHIN EACH ROW (IDENTICAL n , IDENTICAL ATTEMPTED SAMPLES; SKIP PARITY IN NOTE II). MARGIN = WINNER’S ABSOLUTE LEAD; THE WINNER COLUMN IS THE REGIME READ: FASTV’S WINS MARK ITS REFERENCE-GROUNDED HOME REGIME, RBM’S WINS MARK THE OCR REGIME (§V-D). BOLD FULL-SPLIT OCR-BENCH ROWS ARE CONFIRMATION RESULTS AT THE UNIFIED 4 M-PIXEL CAP.

Benchmark (scope; pixel cap)	n	FastV-k3	RBM @25%	Δ (F/R)
Qwen3-VL-8B				
TextVQA (full; default)	5000	0.7771	0.605	+17.2 F
GQA (full; default)	12578	0.5376	0.449	+8.9 F
DocVQA (dev; 600k cap)	200	0.5863	0.5924	+0.6 R [‡]
OCR-Bench (full; 4M cap)	1000	0.413	0.559	+14.6 R
OCR-Bench (dev; native res.)	200	0.415	0.575	+16.0 R
Qwen2.5-VL-7B				
TextVQA (dev; default)	200	0.7467	0.6683	+7.8 F
GQA (dev; default)	200	0.475	0.520	+4.5 R [‡]
DocVQA (dev; 600k cap)	200	0.4852	0.5062	+2.1 R [‡]
OCR-Bench (full; 4M cap)	1000	0.295	0.382	+8.7 R
OCR-Bench (dev; native res.)	200	0.285	0.370	+8.5 R

Table notes. F = FastV; R = RBM; Δ is the winner’s absolute lead in percentage points. (i) Scope and pixels: n is printed per row; DocVQA cells run at a 600k pixel cap on both arms (iso-pixel within this table); the bold full-split OCR-Bench confirmation rows use a 4 M-pixel cap on both arms; the older dev OCR-Bench rows run at native resolution, which is what triggers the skips (note ii); TextVQA/GQA use each model’s default imaging configuration. (ii) OCR-Bench skips: native-resolution giant OCR images overrun the context on $\approx 10\%$ of the $n = 200$ set (19/200 Qwen3-VL, 20/200 Qwen2.5-VL); the *identical* hard samples are skipped under both arms (same paired batch, mode-independent). Table scores use the nominal $n = 200$ denominator and conservatively count skipped samples as zero. All other cells skip ≤ 5 (Qwen3-VL TextVQA FastV, 5/5000) or 0. (iii) Harness: every $n = 200$ cell on both sides comes from the same HF transformers eager-attention harness with family-correct native positions; the Qwen3-VL TextVQA/GQA RBM references are the vLLM full-split cells of Table I, for which engine parity is validated (§IV-A; HF-pre vs vLLM-pre @25% further agree 14/16 with kept-set Jaccard 0.986, §VI-d). An early Qwen2.5-VL $\times$ pre position-harness degeneration (vllm-mimic layout) was detected and replaced by the native layout before any reported number was produced; no reported cell comes from the degenerate path (on Qwen3-VL, native vs mimic differ, 0.575 vs 0.525, same ordering). (iv) Budget: FastV-k3 ≈ 214 vs RBM ≈ 213 mean visual tokens/request (Qwen3-VL TextVQA, 25% retention; the $n = 200$ probe measured 208) — approximately matched, stated per cell; this is “each method at its strong configuration,” not a byte-iso-budget cell. (v) [‡]The Qwen3-VL DocVQA paired difference is inconclusive. For Qwen2.5-VL GQA/DocVQA, the paired 95% CIs cross 0 ($[-10.5, +1.5] / [-10.1, +5.8]$); these cells are exploratory/inconclusive, not statistically robust RBM edges. (vi) The full-split OCR confirmation uses the same HF eager harness and 4 M-pixel cap for both methods on all 1000 items. The nominal scores are 559/1000 vs. 413/1000 for Qwen3-VL (79 symmetric skips; $n_{\text{paired}} = 921$) and 382/1000 vs. 295/1000 for Qwen2.5-VL (88 symmetric skips; $n_{\text{paired}} = 912$); skipped items count as zero. These rows confirm the OCR regime at full scale but do not replace the native-resolution $n = 200$ regime rows.

< 0.332 (TextVQA, text-dense scene) < 0.360 (GQA, object-centric), with the decision-relevant top-25% overlap rising in the same order (Jaccard@25% = $0.180 < 0.243 < 0.278$; full values in Supplementary S1). The merger’s nonlinear projection thus reorders unit saliency almost from scratch ($\rho = 0.14\text{--}0.36$, nowhere near 1); both the full-ranking correlation and the decision-relevant top-25% overlap rise from the most text-dense benchmark to the object-centric one, so the reshuffle is largest on DocVQA and smallest on GQA — yet substantial on every benchmark.

B. M2 — the demoted units are text-stroke units

With $\text{rank_shift} = \text{post_rank} - \text{pre_rank}$ ($+$ $\Rightarrow$ demoted) and per-unit Sobel edge energy as a text-stroke proxy, the merger preferentially *demotes* high-edge units at $\kappa = 25\%$: $\rho(\text{rank_shift}, \text{edge}) = +0.439$ on DocVQA, $+0.155$ on TextVQA, $+0.036$ on GQA — $\sim 12\times$ stronger on documents than on object scenes. The units post drops that pre keeps are the highest-edge of all: on DocVQA, mean Sobel **0.641 vs 0.124** for the reverse group, with **92% above the per-image median (vs 35%)** — post-merger selection is **systematically anti-text on documents**. (The *direction*, we hypothesize, is set by the merger’s training on natural content, which attenuates high-frequency stroke energy; this is a testable hypothesis, not a measured result.) The full M2 table (per-benchmark Sobel means and median fractions) is in Supplementary S2.

C. M3 — a ranking-swap causal control, and the decisive kept-set identity

By unit equivalence (§III-C), holding the *forward path* fixed at post and swapping in the *pre ranking* must reproduce pre accuracy if the gap is purely a ranking effect. It does:

Swapping in the pre ranking recovers full pre-merger accuracy (exactly on DocVQA; within 2/200 greedy-decode answers on TextVQA, ε -level kernel numerics confirmed by a 200/200 byte-identical independent rerun; full values are in Supplementary S9) and erases the collapse ($+38.3$ pp TextVQA, $+26.5$ pp DocVQA over post). The swap path agrees with post on only 40/200 (TextVQA) and 63/200 (DocVQA) answers — without the pre ranking, the same forward path produces the collapse.

The decisive hardening — kept-set identity on both architectures. Emitting the kept unit indices from both paths and comparing directly: **Jaccard(swap-kept, pre-top- κ) = 1.000 in all four cells** (Qwen3-VL and Qwen2.5-VL $\times$ DocVQA and TextVQA; $n = 32$, chance level $\kappa/N = 0.25$). The two paths select *exactly the same units*. The small swap-vs-pre answer residual on Qwen2.5-VL ($+6.7$ pp DocVQA at $n = 32$) traces to an implementation confound of the swap control — window-attention reverse_indices (three recovery sites on Qwen2.5-VL, zero on Qwen3-VL) map identical index sets to physical units in different orderings — while batch-dependent merging is structurally excluded (the merger is per-token LN+MLP). **Verdict: on Qwen3-VL, the tested pre>post gap is attributable to the ranking alone at selection level** (byte-identical); Qwen2.5-VL corroborates kept-set identity but retains an ordering-confound residual. The merged representations of kept units carry no stage-dependent loss on Qwen3-VL (byte-exact); on Qwen2.5-VL the kept *set* is identical but the ordering confound is not eliminated, so we scope the no-loss statement to the byte-exact family.

D. Failure modes and scope boundaries

DocVQA exhibits selection misdirection: post ranking moves away from text and the swap fully restores accuracy. TextVQA combines rank corruption with a budget effect, while GQA has near-null text-stroke directionality ($\rho = 0.04$) but favors

post-stage selection in the corrected matched-boundary control. The stage effect is therefore workload-conditional. It retains sign and magnitude under a global-centroid attention scorer on Qwen3-VL, but that proxy degrades on Qwen2.5-VL; L2 remains the paper’s fixed selector. The ranking law replicates on Qwen2.5-VL and at the accuracy level on InternVL3 (Table I), whereas the causal decomposition is claimed only for byte-exact Qwen3-VL. A low 600k pixel cap reverses shallow Qwen2.5-VL DocVQA; the native-resolution full split remains the headline configuration (§VIII).

Retention curve. The stages are nearly indistinguishable at 75% retention; their gap then widens monotonically through 25% on both Qwen models (full values in Supplementary S3, $n = 200$).

Merger mechanism. Figure 5 connects the stage effect to the native merger. Across four workloads, pre- and post-merger rankings share only moderate rank correlation and top-quartile overlap. Units rich in text-stroke edge energy are preferentially demoted after merging. Finally, a Qwen3-VL ranking-swap control keeps the selected unit set fixed (Jaccard = 1.000) while restoring the pre-merger ordering, separating ranking damage from a change in token identity.

VI. NEGATIVE RESULTS — TESTED EXTENSIONS DO NOT IMPROVE PLAIN RBM

We prespecified five query-aware or regime-aware extensions and report all five as failures on Qwen3-VL-8B. **(a) Image-level routing:** pre is at least as accurate as post on 84–97% of images per benchmark; the best router scores below always-pre (0.484 vs 0.494) and far below the per-sample oracle (0.576). On pooled $N = 774$, only $\sim 27\%$ of the oracle gap is workload-level, leaving $\sim 73\%$ sample- and query-dependent. **(b) Merger-aware masking:** its selected $t = 0.5$ hybrid gains 2.3 pp on TextVQA but loses 8.0 pp on OCR-Bench and gains nothing on GQA; no candidate passes the no-OCR-regression plus GQA-gain gate. **(c) Query-embedding blending:** on a disjoint development slice every $\lambda > 0$ underperforms plain L2 ($-1.7/-5.2/-3.3$ pp for $\lambda = 0.3/0.5/0.7$), so the prespecified rule selects $\lambda = 0$. **(d) A variance-weighted (frequency-aware) selector:** the mechanism-side probe of §III-B ($\alpha=1$, $\beta=0.6$, selected in two grid rounds on TextVQA-500 at 25% retention) improves the text-dense cell (+1.2 pp TextVQA@25%, +0.5 pp at 50%) but regresses -1.6 to -2.2 pp on DocVQA@25%, OCR-Bench@25%, and GQA@25%; no candidate passes a no-regression gate across the four benchmarks, so we freeze the L2 selector and treat the improvement as hypothesis support only. **(e) An adaptive PRE/POST stage router:** a query-blind workload detector on merger-input features (high-frequency tail ratio + L2-entropy) routes each image to the stage predicted to be stronger. Tuned on TextVQA it reduces to always-PRE — statistically indistinguishable from plain RBM on every benchmark (TextVQA/DocVQA/OCR-Bench regression = 0.0 pp, but GQA gain = +0.0 pp, below the required +1 pp); raising the threshold to route scene images to POST collapses TextVQA by a further 7 pp. The GQA and TextVQA workload signatures overlap at the unit level, so no single global threshold reconciles the two acceptance criteria.

Exploratory learned boundary scorer. A held-out MLP probe successfully distills post-merger importance from pre-merger norm, edge, variance, position, and neighborhood features, but transfers the wrong task bias: at 25% retention, TextVQA falls from 0.595 to 0.435 (-16.0 pp) while GQA rises from 0.515 to 0.550 (+3.5 pp). It is therefore not an RBM upgrade; as an intervention, its failure is consistent with M2’s text-stroke demotion (Supplementary S4b.3).

These results bound, rather than close, the tested design space: image-only signals cannot access the query-dependent oracle headroom, and distilling a post-merger teacher reproduces its text-dense failure mode. We therefore freeze plain RBM as the robust text-dense/OCR default and select a FastV-class method for scene-text/object-centric traffic; within the tested configurations, routing, scorer modification, and learned post-order imitation offer no consistent benefit.

VII. EFFICIENCY — SUPPORTING EVIDENCE

Efficiency is supporting evidence, measured only in vLLM 0.19 V1 offline on one A40. **Stage-neutral:** across 75/50/25% retention, pre vs post throughput is 4.61/4.52, 5.39/5.54, and 6.91/6.89 req/s; five-repeat measurements remain within 3% with no consistent direction. **Compression scaling:** relative to no compression, throughput rises 12/31/68% and 25%-retention latency falls from 0.36 to 0.23 s. InternVL3 independently yields $1.8\text{--}2.5\times$ throughput at 25% retention, again with pre $\approx$ post at identical token counts. The flat ≈ 42 GB allocator reservation is not a memory measurement; continuous-batching efficiency remains future work.

VIII. DISCUSSION AND LIMITATIONS

The broader design implication is to determine the evidence set before a lossy native aggregation step when that step is not rank-preserving, then reuse the model’s unmodified merger on the survivors. Within the tested merger-equipped models and scorer families, RBM instantiates this rule and better preserves text-dense accuracy; Qwen3-VL diagnostics indicate that its native merger rewrites saliency in a text-hostile direction. This is a bounded design rule, not a universal winner. RBM retains dense OCR better than the compared FastV settings (+16.0/+8.5 pp on the two Qwen models), while FastV leads on several scene-text/object-centric cells (Table II).

Architecture and mechanism scope. The byte-exact M3 causal attribution is established on Qwen3-VL; Qwen2.5-VL retains an ordering confound, and InternVL3 provides accuracy-level and bounded ranking-swap support rather than a byte-exact causal decomposition. Models without a native merger are out of scope. M1/M2 use a Sobel proxy on $n = 64$ samples per benchmark and M3 uses $n = 200$; the full-split GQA control favors post by 5.64 pp at the matched boundary, and no equivalence margin was preregistered. Holm correction covers the nine prespecified 25%-retention text-dense stage contrasts; secondary budgets, GQA, mechanism analyses, baseline regime comparisons, and extension gates remain uncorrected and should be read as secondary or exploratory.

Evaluation scope. FastV uses HF eager whereas the other methods use vLLM; anchors support accuracy comparison, but

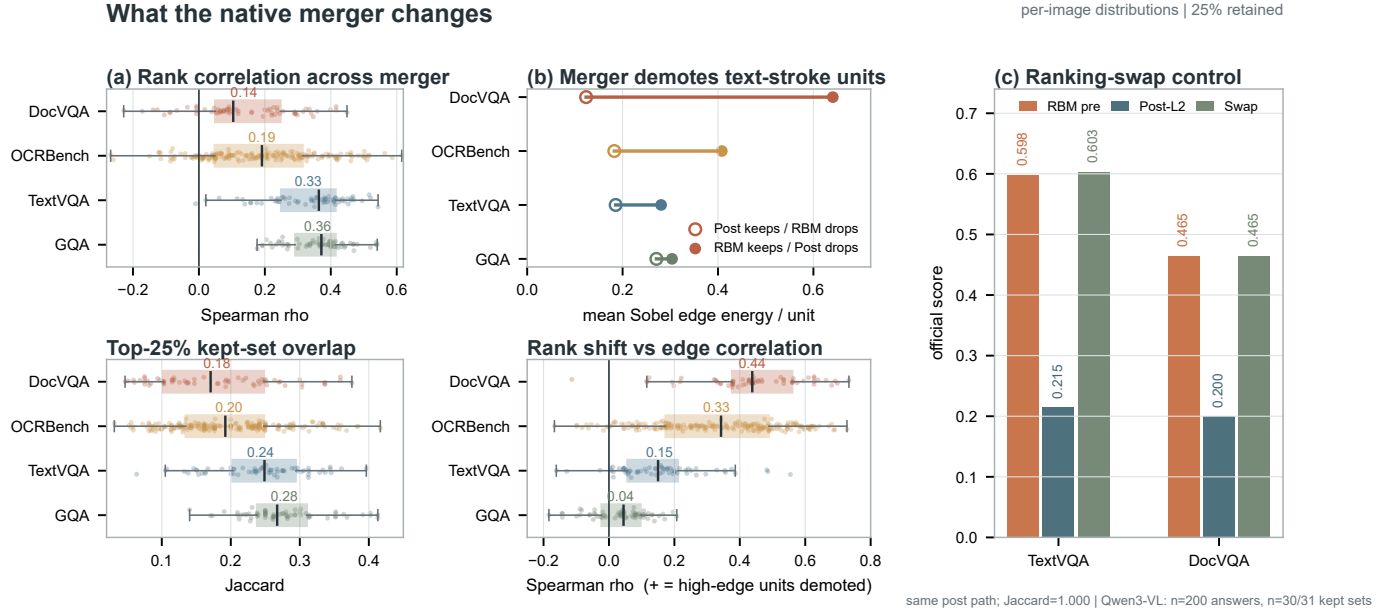


Fig. 5. **What the native merger changes.** (a) Pre/post L2 rankings are only moderately preserved, measured by Spearman correlation and top-25% kept-set Jaccard across DocVQA ($n = 64$), OCRBench ($n = 199$), TextVQA ($n = 64$), and GQA ($n = 64$). (b) Rank shifts are associated with text-stroke edge energy: units with stronger high-frequency structure are demoted more after merging. (c) In the Qwen3-VL ranking-swap control, post-merger features use the pre-merger order while the kept set remains identical (Jaccard = 1.000), isolating ordering from token identity on $n = 200$ TextVQA and DocVQA samples.

throughput is vLLM-only. Some baseline cells use $n = 200$, as marked. Metric-level bootstrap intervals and paired permutation tests are primary; McNemar covers additional binary outcomes.

Configuration scope. DocVQA pixel caps differ (4 M for InternVL3, 600k for Qwen baseline cells), so cross-model absolute scores are incomparable. Low-resolution Qwen2.5-VL reverses the shallow-stage direction; the headline is the native-resolution full split. RBM also remains lossy against no compression on InternVL3 DocVQA/OCR (-19.4 pp/ -99 points), despite beating post.

IX. CONCLUSION

We formulate visual-token compression around native merging as a representation-dependent selection-order problem and instantiate a stage-consistent solution, RBM, under an explicit unit-and-interface contract. Under iso-model, iso-budget, and a shared query-blind L2-magnitude family, pre-merger ranking is consistently stronger across the tested text-dense cells of three models from two architecture families; the measured gaps generally grow under deeper compression. A ranking-swap control attributes the Qwen3-VL gap to merger-induced rank changes. The resulting training-free method, RBM, is a robust OCR-oriented default rather than a universal winner: query-conditioned FastV remains stronger on several scene-text/object-centric settings. RBM adds no stage-specific serving cost, and five prespecified Qwen3-VL extensions fail to improve over their stronger constituents.

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